
Education

PhD in Biostatistics

May 2018

Johns Hopkins Bloomberg School of Public Health

Dissertation: Evidence-Based Methods in Studies of Biology and Data Analysis

Advisor: Kasper Daniel Hansen

BS in Biomedical Engineering

May 2013

Johns Hopkins University

Secondary major: Applied Mathematics and Statistics

Minor: Computer Science

Work Experience

Associate Professor

2023 - present

Assistant Professor

2018 - 2023

Department of Mathematics, Statistics, and Computer Science

Macalester College, Saint Paul, MN

Johns Hopkins Biostatistics Center

July 2016 - August 2017

Student statistical consultant

Johns Hopkins Bloomberg School of Public Health, Baltimore, MD

Advisor: Carol Thompson, MS

Published

12. Al Awadhi, S., **Myint, L***, Guallar, E., Clish, C. B., Wulczyn, K. E., Kalim, S., Thandhani, R., Segev, D. L., McAdams-DeMarco, M., Moe, S. M., Moorthi, R. N., Himmelfarb, J., Powe, N. R., Tonelli, M., Rhee, E. P., & Shafi, T. (2025). A metabolomics approach to identify metabolites associated with uremic symptoms in patients receiving maintenance hemodialysis. *Kidney360*, 10.34067/KID.0000000838. <https://doi.org/10.34067/KID.0000000838>

* Indicates co-first authorship

11. Al Awadhi, S., **Myint, L***, Guallar, E., Clish, C. B., Wulczyn, K. E., Kalim, S., Thadhani, R., Segev, D. L., McAdams DeMarco, M., Moe, S. M., Moorthi, R. N., Hostetter, T. H., Himmelfarb, J., Meyer, T. W., Powe, N. R., Tonelli, M., Rhee, E. P., & Shafi, T. (2024). A metabolomics approach to identify metabolites associated with mortality in patients receiving maintenance hemodialysis. *Kidney International Reports*, 0(0). <https://doi.org/10.1016/j.ekir.2024.06.039>

* Indicates co-first authorship

10. **Myint, L**. 2023. Controlling time-varying confounding in difference-in-differences studies using the time-varying treatments framework. *Health Services & Outcomes Research Methodology*. <https://doi.org/10.1007/s10742-023-00305-2>
9. Wulczyn, K., Rhee, E., **Myint, L**, Kalim, S., & Shafi, T. 2022. Incidence and Risk Factors for Pruritus in Patients with Nondialysis CKD. *Clinical Journal of the American Society of Nephrology: CJASN*. <https://doi.org/10.2215/CJN.09480822>
8. Hu J-R, **Myint L***, Levey AS, Coresh J, Inker LA, Grams ME, Guallar E, Hansen KD, Rhee EP, Shafi T. 2021. A metabolomics approach identified toxins associated with uremic symptoms in advanced chronic kidney disease. *Kidney international*. DOI: 10.1016/j.kint.2021.10.035.
- * Indicates co-first authorship
7. **Myint L**, Hadavand A, Jager L, Leek J. 2019. Comparison of Beginning R Students' Perceptions of Peer-Made Plots Created in Two Plotting Systems: A Randomized Experiment. *Journal of Statistics Education* 28:98–108. DOI: 10.1080/10691898.2019.1695554.
6. **Myint L**, Wang R, Boukas L, Hansen KD, Goff LA, Avramopoulos D. 2019. A screen of 1,049 schizophrenia and 30 Alzheimer's-associated variants for regulatory potential. *American Journal of Medical Genetics Part B: Neuropsychiatric Genetics* 183:61–73. DOI: 10.1002/ajmg.b.32761.
5. **Myint L**, Avramopoulos DG, Goff LA, Hansen KD. 2019. Linear models enable powerful differential activity analysis in massively parallel reporter assays. *BMC Genomics* 20:209. DOI: 10.1186/s12864-019-5556-x.
4. **Myint L**, Leek JT, Jager LR. 2018. Explanation of observational data engenders a causal belief about smoking and cancer. *PeerJ* 6:e5597. DOI: 10.7717/peerj.5597.

- **Press:** Preprint was featured in the [February 2018 issue](#) of Significance Magazine.

3. Monroe AK, **Myint L**, M Rutstein R, Aberg J, L Boswell S, L Agwu A, A Gebo K, D Moore R, HIV Research Network. 2018. Factors Associated With Gaps in Medicaid Enrollment Among People With HIV and the Effect of Gaps on Viral Suppression. *Journal of Acquired Immune*

Deficiency Syndromes 78:413–420. DOI: 10.1097/QAI.0000000000001702.

2. Kang JY, Rabiei AH, **Myint L**, Nei M. 2017. Equivocal significance of post-ictal generalized EEG suppression as a marker of SUDEP risk. *Seizure: The Journal of the British Epilepsy Association* 48:28–32. DOI: 10.1016/j.seizure.2017.03.017.
1. **Myint L**, Kleensang A, Zhao L, Hartung T, Hansen KD. 2017. Joint Bounding of Peaks Across Samples Improves Differential Analysis in Mass Spectrometry-Based Metabolomics. *Analytical Chemistry* 89:3517–3523. DOI: 10.1021/acs.analchem.6b04719.

Non-Reviewed Scholarship

- Heggseth B, **Myint L**, Grinde K. (2018) "STAT 155 Notes" Course text for Introduction to Statistical Modeling. Published at <https://mac-stat.github.io/Stat155Notes/>
 - Grinde K, Theobald A, **Myint L**. [Beyond Achievement: Access, Identity, and Power in Alternative Grading](#). May 2024. Guest post for the [Grading for Growth blog](#).
 - **Press:** On May 27, 2024, this post was [featured](#) as the 4th most opened post of all time.
 - **Myint L**. [Who is the teacher that teaches?](#) July 2024. Guest post for the [AMS Column on Teaching and Learning](#).
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Software

yamss: Tools for the analysis of high-throughput metabolomics data. An R package released through the Bioconductor project.

<https://www.bioconductor.org/packages/yamss>

mpira: Tools for the analysis of data from massively parallel reporter assays. An R package released through the Bioconductor project.

<https://www.bioconductor.org/packages/mpira>

rsemmed: A programmatic interface to the Semantic MEDLINE database. A tool for computational literature discovery.

<https://www.bioconductor.org/packages/rsemmed>

Grant Participation

Metabolomics of Uremic Symptoms in Dialysis Patients

NIH R01 subaward contract

Dates: June 2019 - January 2025

Role: statistical analyst

Presentations

Adventure in Alternative Grading

Spring Professional Activities Workshop. Macalester College. ([slides](#))
May 2024

Access as the intersection of resources and practices

Joint Statistical Meetings. [Invited session](#). ([slides](#))
August 2023

Controlling time-varying confounding in difference-in-differences studies using the time-varying treatments framework

American Causal Inference Conference ([poster](#))
May 2023

Alternative grading for "rigor" and beauty

Macalester College - Center for Scholarship and Teaching seminar ([slides](#))
April 2023

Causal inference throughout the statistics curriculum

Joint Statistical Meetings. [Invited session](#). ([slides](#))
August 2022

A metabolic view of symptoms in kidney disease

University of St. Thomas - Department of Applied Probability and Statistics seminar ([slides](#))
April 2022

Exploring biomedical concept graphs with rsemmed

Bioconductor Conference 2021 ([poster](#))
August 2021

What Did I Just Read? Organizing Knowledge From the Research Literature Using Graph Databases

Minnesota Women in Analytics and Data Science conference ([slides](#))
October 2020

Graphs Galore! Representing Knowledge in the Sciences and Humanities

Carleton College - Department of Statistics seminar ([slides](#))
October 2019
Macalester College - Center for Scholarship and Teaching seminar ([slides](#))
November 2019

Statistical methods for querying the regulatory role of DNA

Creighton University Math Colloquium ([slides](#))
March 2019

Magical Web Scraping with rvest

Baltimore R Ladies Group ([slides](#))

May 2018

Joint Preprocessing of Samples Improves Power in Differential Analysis for Mass Spectrometry-Based Metabolomics

Johns Hopkins University - Department of Biophysics seminar

December 2017

Shiny Applications for Teaching and Dungeons and Dragons

Baltimore UseR Group ([slides](#))

September 2017

A Method for Joint Processing of Mass Spectrometry-Based Metabolomics Data for Improved Differential Analysis

Poster: ENAR, Washington D.C.

March 2017

Macalester College

Instructor

- COMP/STAT 112: Introduction to Data Science (S26)
- STAT 125: Epidemiology (F19, S24)
- STAT 155: Introduction to Statistical Modeling (F18, S20, F20, F24)
- COMP/STAT 212: Intermediate Data Science (F23, S24)
- STAT 253: Statistical Machine Learning (S19, S21, S23, S26)
- STAT 451: Causal Inference (F20, S23, F24) (Formerly STAT 394 (S20))

Cloud-Based Data Science Specialization

Content developer

A massive open online course on the Leanpub platform for providing a highly accessible data science education. Content developer for the following courses:

- [Organizing Data Science Projects](#)
- [Version Control](#)
- [Introduction to R](#)
- [Data Tidying](#)

Johns Hopkins Bloomberg School of Public Health

Instructor

- Statistical Thinking for Informed Decision Making (2 semesters)
I developed this course as part of the [Gordis Teaching Fellowship](#), a school-wide award that provides funds to design and teach an undergraduate class. A news article-motivated introduction to major biostatistical areas, including causal inference, survey sampling, and survival analysis.

Teaching Assistant

- Public Health Biostatistics (3 semesters)
- Introduction to R for Public Health Researchers (1 course)
- Statistical Methods in Public Health (3 quarters)
- Data Analysis Workshop (2 courses)
- Statistics for Genomics (1 quarter)
- Statistics for Laboratory Scientists (2 quarters)
- Summer Institute: Statistical Reasoning in Public Health (2 courses)

Tutor

- Statistical Methods in Public Health (2 quarters)
- Mentor for Center for Talented Youth Cogito Research Award Recipient (3 months)

Johns Hopkins University

Teaching Assistant

- Introduction to Java (1 semester)

Awards

Helen Abbey Award

May 2017

Johns Hopkins Bloomberg School of Public Health

Excellence in teaching ([website](#))

Professional Service

Memberships

- American Statistical Association

Conferences

- 2020: Organizer and Chair for an invited session for the [Education Track](#) at the Symposium on Data Science and Statistics

Reviewer

- 2023: Health Services and Outcomes Research Methodology
- 2023: Journal of Statistics and Data Science Education
- 2022: Causal Learning and Reasoning (CLear 2023) conference
- 2022: Clinical Journal of the American Society of Nephrology
- 2021: PeerJ
- 2020: [Open Case Studies](#)
- 2020: Journal of Statistics Education
- 2019: Nature Human Behaviour
- 2019: Technology Innovations in Statistics Education
- 2019: Journal of Statistics Education
- 2018: [BiOverlay](#)
- 2018: American Journal of Epidemiology
- 2017: Observational Studies

Institutional Service

Department Service

- MSCS Transformation Committee (2024-5)
- MSCS Honors Seminar Organizing Committee (2023-4)
- Mathematics Tenure-Track Position Hiring Committee (2023)
- DataFest Organizer (2023)
- Student Community Committee (2023)
- Revision of Department Majors (2021)
- Statistics Tenure-Track Position Hiring Committee (2019, 2022, 2025)
- Statistics Allocation Request (2019, 2022)
- Statistics Visitor Hiring Committee (2020, 2023)
- Statistics Postdoc Hiring Committee (2018, 2020)

Undergraduate Research Advising

Honors Advisor

- Alicia Severiano Perez (2025-2026)
- Wenxuan Zhu (2023-2024)
- Ben Christensen (2023-2024)
- Charlotte Giang (2020-2021)
- Fiona Adams (2019-2020)

Honors Reader

- Giovanny Martínez Rodríguez (2025)
- Paige Tomer (2024)
- Sarah Tannert-Lerner (2023)
- Connie Zhang (2023)
- Zuofu Huang (2021)
- Liam Purkey (2021)
- Coco Li (2021)
- Aidan Toner-Rodgers (2021)
- Millie Baker (2020)
- Kieu-Giang Nguyen (2020)
- Katie Jolly (2019)

Student Research Supervisor

- Josh Rollin (Summer 2023)
- Ben Christensen (Summer 2023)
- Nicholas Di (Summer 2019)

Internship Supervisor

- Juliet Cramer (Spring 2024)
- Ronan Manning (Spring 2024)

College Service

- *All We Can Save* reading group organizer (2025-2026)
- Sustainability Advisory Committee (2025-present)
- Community and Global Health Concentration Co-Director (2023-2024)
- Strategic Planning Implementation: First-Year Experience Committee (2023-2024)
- Strategic Planning Implementation: Student-Ready Campus Committee (2023)
- COVID Testing Committee (2020-2021)
- Community and Global Health Concentration Steering Committee (2020-present)